



Resume Screening & Application Tracking System

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Abstract

The Resume Screening & Application Tracking System is a lightweight, web-based recruitment management application implemented with HTML5, CSS3 and JavaScript (ES6). The system automates candidate data management, performs skill-based resume screening, computes an automatic match-percentage score, and provides status tracking and dashboard analytics. Persisting data in browser local storage enables a portable, zero-backend deployment on GitHub Pages.

The primary purpose is to reduce manual effort and time-to-hire by offering rapid, reproducible screening and simple reporting. Expected outcomes include improved shortlisting accuracy, consolidated candidate records, and exportable recruitment reports suitable for evaluative review by hiring teams and faculty examiners.

Introduction



Context

Recruitment systems coordinate the lifecycle of hiring from vacancy posting to onboarding. Modern organisations rely on structured processes to handle volumes of applicants while preserving fairness and traceability.



Importance of Screening

Resume screening filters applicants by skills and experience, ensuring that interview resources focus on candidates with requisite qualifications. Effective screening improves match quality and reduces recruiter workload.



Need for Automation

Manual hiring workflows are slow and error-prone. Automation mitigates bias, accelerates decision cycles, and scales with applicant volume—key advantages for departments and industry partners evaluating technical candidates.

Problem Statement

Manual resume screening and spreadsheet-based tracking present several limitations: inefficient use of recruiter time, inconsistent evaluation criteria, and difficulty maintaining a searchable, centralised applicant database. Large applicant pools exacerbate these issues, leading to delayed hiring decisions and loss of qualified candidates.

There is a clear need for a simple, reliable system that automates skill matching, maintains candidate state across the recruitment pipeline, and produces concise analytics and reports to support evidence-based decision-making.

Objectives

Automate candidate screening

Implement algorithmic skill matching and automatic match-percentage calculation.

Improve recruitment efficiency

Streamline candidate management to reduce time-to-shortlist.

Reduce manual effort

Provide CRUD operations and local persistence to replace ad-hoc spreadsheets.

Provide recruitment insights

Dashboard analytics and success-rate calculations for evaluative reporting.

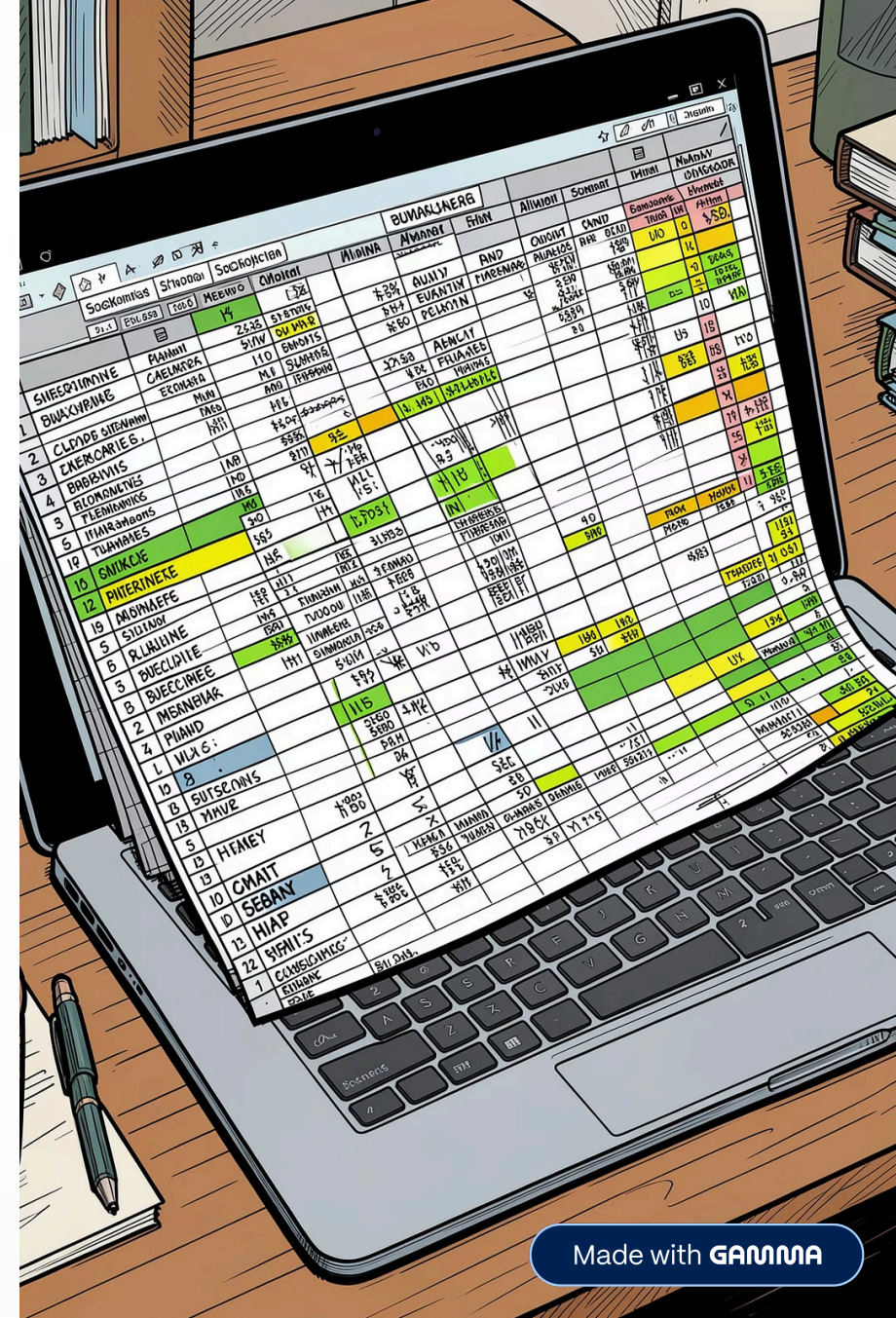
Enable status tracking

Maintain granular candidate states (Applied, Shortlisted, Rejected) through the pipeline.

Existing System

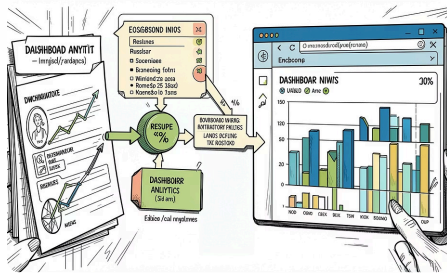
Conventional recruitment in many small organisations relies on manual collection and spreadsheet management. Recruiters copy-paste candidate information, track status columns, and perform ad-hoc filters. This approach is brittle—susceptible to human error, lacks auditability, and performs poorly under high applicant volume.

Spreadsheet-driven processes provide limited automation for skill matching and require repeated manual aggregation to generate reports. They also lack web accessibility and responsive interfaces needed for modern evaluation workflows.



Proposed System

The proposed Resume Screening & Application Tracking System is a responsive single-page web application built with HTML5, CSS3 and JavaScript. It stores candidate records in browser local storage and runs entirely on the client, enabling immediate deployment via GitHub Pages.



Advantages

Lightweight deployment, intuitive UI, automated match scoring, quick search and filtering, and downloadable recruitment reports—delivering efficiency gains over manual workflows.



Workflow

Candidates are added via a form → data persisted locally → resumes are screened against required skills → match percentage and status are computed → dashboard visualises results → reports exported.

System Architecture



The system is modular and client-centric. The Candidate Management Module provides CRUD operations. The Resume Screening Module tokenises skill sets, compares them to required skills, and computes a match-percentage. The Dashboard Analytics Module aggregates status counts and success rates. The Report Generation Module compiles visible data into a downloadable file. The Search and Filtering Module supports queries by name, skills and status. All modules interact with the browser's local storage as the persistence layer and the entire application is delivered via GitHub Pages.

Technologies Used

HTML5

Semantic structure and accessible forms enable robust data entry and a responsive layout. HTML5 provides input types and attributes that simplify validation and improve usability.

CSS3

Cascading Style Sheets supply layout control, responsive grid and flexbox patterns, and polished visual hierarchy. Theme colours (#505468ff, #F4F4F5, #E2E3E9) are applied for coherence and accessibility.



JavaScript (ES6)

Client-side scripting implements application logic: DOM manipulation, event handling, match-percentage calculation, and local storage management. The project uses modular functions and modern ES6 features for clarity and maintainability.



Local Storage

Browser `localStorage` stores JSON-serialised candidate records, enabling persistence across sessions without server infrastructure. This simplifies deployment while retaining basic data durability for single-user scenarios.

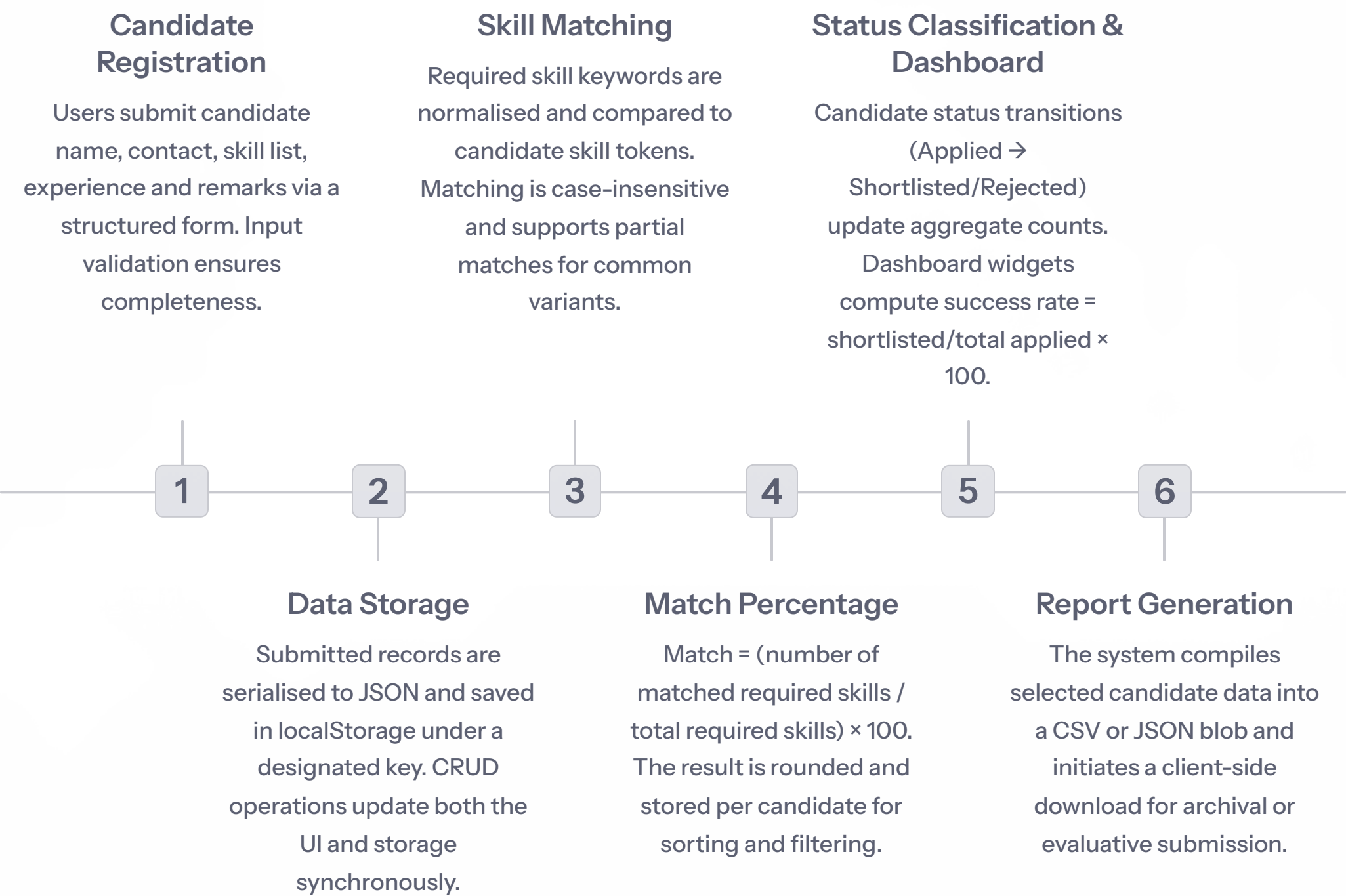


GitHub Pages

Hosting the static site on GitHub Pages provides a free, reliable delivery mechanism for the client-side application.

Continuous deployment is achieved by committing the build to the repository's pages branch.

Methodology, Implementation & UI



Implementation Details (Feature Summary)

- Add Candidate — validated form, adds record to local storage and refreshes table.
- Edit Candidate — modal or inline editing updates storage and recalculates match if skills changed.
- Delete Candidate — removes record with confirmation, updates analytics.
- Search Candidate — text-based search across name and skills, real-time filtering.
- Resume Screening — tokenisation and keyword matching to compute scores.
- Match Logic — exact and partial keyword matching with normalization; configurable required skills per vacancy.
- Status Tracking — enumerated states with timestamps for auditability.
- Dashboard Analytics — visual counts, match distributions and success rate calculation.
- Recommendation Engine — threshold-based recommendations (e.g., $\text{match} \geq 70\% \rightarrow \text{recommend shortlist}$).
- Download Report — CSV/JSON export of filtered view, triggered client-side.